Letter combinations of a phone number

Given a string containing digits from 2-9 inclusive, return all possible letter combinations that the number could represent. Return the answer in **any order**.

A mapping of digits to letters (just like on the telephone buttons) is given below. Note that 1 does not map to any letters.

Example 1:

```
Input: digits = "23"
Output: ["ad","ae","af","bd","be","bf","cd","ce","cf"]
```

Example 2:

```
Input: digits = "2"
Output: ["a","b","c"]
```

input → 2 → a, b, c

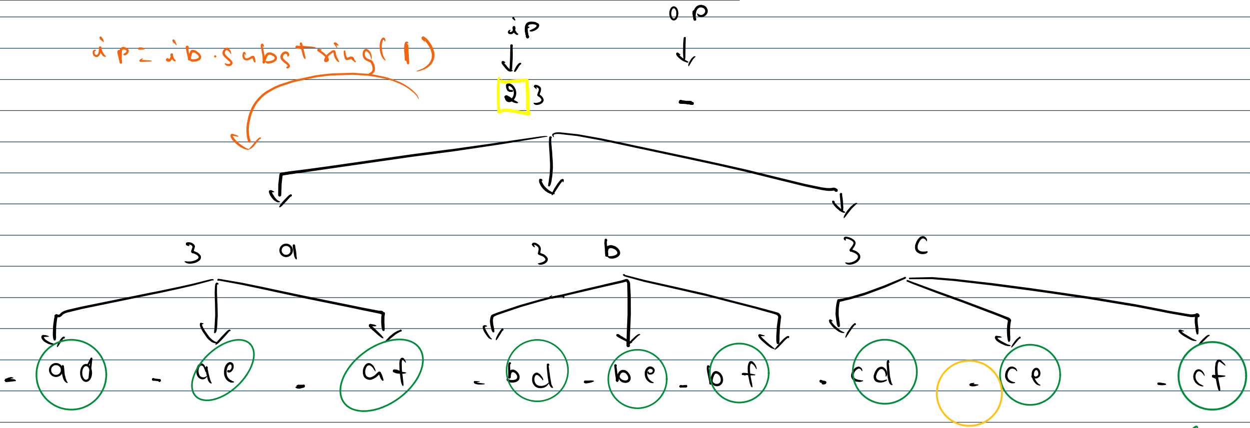
input → 2 3

Choices + Decision } → Recursion

Key's →

""	""	"abc"	"def"	"ghi"	"jkl"	"mno"	"pqrs"	"tuv"	"
0	1	2	3	4	5	6	7	8	

ip = ip.substring(1)



ip.len == 0 → Base condn

op

maximum Product subarray

Given an integer array nums, find a subarray that has the largest product, and return *the product*.

The test cases are generated so that the answer will fit in a **32-bit** integer.

Note that the product of an array with a single element is the value of that element.

Example 1:

```
Input: nums = [2,3,-2,4]
Output: 6
Explanation: [2,3] has the largest product 6.
```

Example 2:

```
Input: nums = [-2,0,-1]
Output: 0
Explanation: The result cannot be 2, because [-2,-1] is not a subarray.
```

numbers → positive ⇒ natural no's } ⇒ whole no's
→ 0
→ negative

nums = [1, 5, 7, 8, 9] ⇒ 1 x 5 x 7 x 8 x 9 = 2520 {+ve no's}

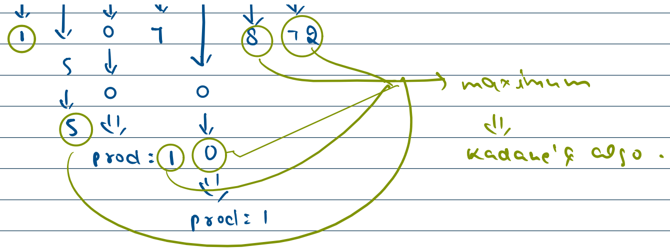
nums = [1, 5, 0, 7, 0, 8, 9] ⇒ 72 {+ve no's, zero's}

1 5 0 7 0 8 72 → maximum

String[]

$\boxed{\text{wxyz"}}$

01



arr → [2, 3, -2, 4]   (indices 0 1 2 3)

cumax : arr[0] : ~~2~~ ~~6~~ -2
cumin : arr[0] : ~~2~~ ~~3~~ -12

max(arr[i], cumax * arr[i], cumin * arr[i])   max : (3, 2* 3, 2* 3) : 6 → Answer..

min(arr[i], cumax * arr[i], cumin * arr[i])   min : (3, 2* 3, 2* 3) : 3

-2 => max(-2, -2*6, -2*3) = (-2, -12, -6) : -2
min(-2, -2*6, -2*3) = (-2, -12, -6) = -12

4 => max(4, -2*4, -12*4) = (4, -8, -48) = 4
= -48

arr → [-2, 3, -4]   (indices 0 1 2) = -2 × 3 × -4 = 24

cu → (arr[i], cumax * arr[i], cumin * arr[i])
↳ min/max

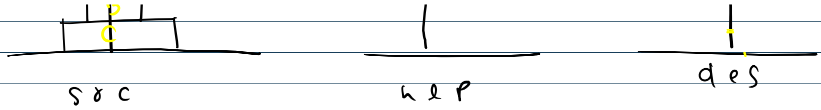
cumin : ~~-2~~ ~~3~~ 24
cumax : ~~-2~~ ~~-6~~ -24

3 => cumax = max(3, -2* 3, -2* 3) = 3
min(3, -6, -6) = -6

Answer

-4 => cumax = max(-4, 3*-4, -6*-4) = 24
(-4, -12, 24)
cumin : min(-4, -12, -24) = -24

Tower of Hanoi Problem



Rules - one disk at a time
Larger disk cannot be placed at smaller disk